M. Omar Nawaz

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PROFILE

I am a research group leader and educator in air quality science working to understand how human activity and societal development influence the climate, degrade air quality and harm public health. My programme integrates interdisciplinary perspectives to explore how big data and AI can inform environmental exposures, health impacts and policymaking, with the aims of improving population health and reducing global exposure inequalities. In service of these aims, I develop health impact and exposure assessment frameworks built on a toolkit of high performance computing, chemical transport modelling, satellite remote sensing, machine learning and data science.

Since 2023 I have secured £540,000 in research grants and commissioned research, including a Wellcome Trust award as Institutional PI, and published 28 peer-reviewed articles (nine as first author, more than 1,000 citations), most recently in *Nature Communications* and *Nature Health*. At Cardiff I lead a group of a postdoctoral researcher, two PhD students and three masters students, and serve as the School's Equality, Diversity, Inclusion and Anti-Racism Lead.

ACADEMIC APPOINTMENTS

Cardiff University, Cardiff, UK

Lecturer in Climate Change Science, Department of Earth and Environmental Sciences — 2025–present

George Washington University, Washington, DC, USA

Postdoctoral Research Associate, Milken Institute School of Public Health — 2023–2025
Professorial Lecturer, Milken Institute School of Public Health — 2024
NRDC Health Science Policy Fellow, Milken Institute School of Public Health — 2023–2024

University of Colorado Boulder, Boulder, CO, USA

PhD Research Assistant, Department of Mechanical Engineering — 2018–2023

University of North Carolina at Chapel Hill, Chapel Hill, NC, USA

MSc Research Assistant, Gillings School of Global Public Health — 2016–2018

EDUCATION

PhD, Air Quality, Department of Mechanical Engineering, CU Boulder — 2023
Thesis: *An adjoint sensitivity framework for public health: the sources of air pollution and their current and future impacts at the urban and national scale*
Supervisor: Professor Daven K. Henze

MSc, Env. Engineering, Gillings School of Global Public Health, UNC Chapel Hill — 2018
Dissertation: *Benefits of reduced premature mortality from decreases in $PM_{2.5}$ and ozone in the United States from 1999 to 2015*
Supervisor: Professor J. Jason West

BSc, Physics, Department of Physics and Astronomy, UNC Chapel Hill — 2017

BSc, Applied Mathematics, Department of Mathematics, UNC Chapel Hill — 2017

RESEARCH FUNDING

I maintain an active record of applying for, winning and leading funded research projects supported by national and regional bodies, charities and industry, totalling £540,000 since 2023, alongside national high-performance computing allocations of 75,000 node-hours from UKRI and 100,000 CPU-hours from Supercomputing Wales.

Research grants (funded role)

Wellcome Trust, **Institutional PI**, *Maximizing health benefits from reducing oil and gas emissions in Mexico*. £496,000* 2025–2026

UKRI, **PI**, *Forward and adjoint modeling of transboundary secondary $PM_{2.5}$ in response to climate mitigation*. 75,000 node-hours 2026–2027

Supercomputing Wales, **PI**, *Estimating UK surface-level pollution from satellite data using machine learning and deterministic modelling*. 100,000 CPU-hours 2025–2026

GW4, **Co-I**, *Mapping Displacement Futures (GW4 Centre on Displacement)*. £2,506 2026–2027

Research grants (unfunded role)

Wellcome Trust, **Technical Panel Member**, *Comprehensive Assessment of Co-Benefits and Maladaptation of Solarisation in Thailand*. £1,339,120 2026–2029

NASA, **Collaborator**, *Application of satellite observations in estimating NO_2 concentrations, mortality burdens, and inequities*. £112,000* 2024–2027

Consultancy and contract research

ICCT, **Consultant**, *Health impact assessment software development*. £13,000 2026

ICCT, **Consultant**, *The health impacts of vehicle emissions in Mexico*. £2,000 2026

ICCT, **Consultant**, *Health benefits of future Colombian transport*. £2,000 2026

ICCT, **Consultant**, *Health benefits of zero emission vehicles by 2050*. £12,000 2025–2026

ICCT, **Consultant**, *Global maritime health impacts and policy benefits*. £5,000 2024

ICCT, **Consultant**, *Study of global transportation health impacts*. £9,000* 2023–2025

Under review

UKRI, **PI**, *Mitigating and adapting to wildfire smoke at the wildland-urban interface under future climate and development scenarios*. £1,673,000* 2027–2030

NERC, **PI**, *Quantifying the cost of the measurement disparity of geostationary air pollution remote sensing in Global South cities*. £55,368 2027–2028

Competitive applications shortlisted

Wellcome Trust Career Development Award, **PI**, *ACCEPT: Assessing Climate Change effects on Equity and Public health from Transboundary air pollution*. £2,110,000 2026

*Converted to GBP and rounded to the nearest thousand. ICCT: International Council on Clean Transportation.

HONOURS AND FELLOWSHIPS

Fellow of the Higher Education Academy (FHEA) 2026

GW4 Crucible cohort, research leadership programme of the GW4 Alliance 2026

Cardiff University Enriching Student Life Award, nominee 2026

GeoCAFE Scholar 2024–2025

Natural Resources Defense Council Health Science Policy Fellowship 2023–2024

Outstanding Mechanical Engineering Research Potential Fellowship, CU Boulder 2018

First place, student poster award, UNC 5th Climate Change Symposium 2018

SELECTED PUBLICATIONS

Over the last six years I have published 28 peer-reviewed journal articles (more than 1,000 citations), nine as first author, and have eight further manuscripts under review. Reflecting the interdisciplinarity of my research programme, I publish in a variety of journals, including general medical journals such as *The Lancet*, multidisciplinary journals such as *Nature Communications*, and population and public health journals such as *Nature Health*. Five key publications are given below with a brief note on their significance; the full list is in the appendix.

1. **Nawaz, M. O.**, Henze, D. K. (2026). National climate action can ameliorate, perpetuate, or exacerbate international air pollution inequalities. *Nature Communications*, doi:10.1038/s41467-026-68827-0.
   In this first-author publication I developed a health modelling framework, based on adjoint sensitivity analysis, to quantify inequality in how air pollution health impacts are exchanged between countries under different climate futures, and assessed the resulting health disparities between the Global North and South.

2. Hay, S. I., Ong, K. L., Santomauro, D. F., … **Nawaz, M. O.**, … et al. (2025). Burden of 375 diseases and injuries, risk-attributable burden of 88 risk factors, and healthy life expectancy in 204 countries and territories, including 660 subnational locations, 1990–2023: a systematic analysis for the Global Burden of Disease Study 2023. *The Lancet*, 406, 1873–1922.
   For the Global Burden of Disease Study 2023 I developed the exposure dataset for nitrogen dioxide (NO_2) air pollution and worked alongside other public health experts to assess the years lived with disability from childhood asthma globally from 1990 to the present.

3. **Nawaz, M. O.**, Goldberg, D. L., Kerr, G. H., Anenberg, S. C. (2025). TROPOMI satellite data reshape NO_2 air pollution land-use regression modeling capabilities in the United States. *ACS ES&T Air*, 2, 187–200.
   In this first-author publication I developed a land-use regression model of NO_2 exposure for the contiguous United States using TROPOMI satellite observations as predictors. I am building on this work with NGOs and other researchers to assess lung cancer mortality and environmental injustice in exposure.

4. **Nawaz, M. O.**, Henze, D. K., Anenberg, S. C., Braun, C., Miller, J., Pronk, E. (2023). A source apportionment and emission scenario assessment of $PM_{2.5}$- and O_3-related health impacts in G20 countries. *GeoHealth*, 7, e2022GH000713.
   In this first-author publication I contrast how domestic and transboundary sectoral sources differ in their contributions to national air pollution health burdens in G20 countries, and project the public health co-benefits of achieving net-zero CO_2 targets.

5. Kerr, G. H., **Nawaz, M. O.**, Anenberg, S. C., Anthoff, D., Burton, C., Carter, T. S., Henze, D. K., Kelley, D. I., Kingdon, C., O'Dell, K., Prest, B. C., Cromar, K. R. (2026). Climate-driven surges in public health damages from wildland fire-sourced pollution through 2100. *Nature Health*, accepted.
   This second-author publication projects the public health damages from wildland fire pollution under climate change through 2100. Building off the approach developed in my 2020 first-author paper, we combine adjoint sensitivities with projected changes in biomass burning emissions to characterise public health impacts.

INVITED TALKS

1.	*The added benefit of TEMPO remote sensing of NO_2 for exposure and health impact assessment.* CEOS Atmospheric Composition Virtual Constellation, Leiden, Netherlands, 1–5 June 2026.	2026
2.	*Using HPC for the variable resolution simulation of air pollution formation using the GCHP stretched-grid model.* Advanced Research Computing Operational and User Group, Cardiff, UK, 20 May 2026.	2026
3.	*Applying machine learning and statistical modeling approaches to remote sensing observations for inferring surface-level NO_2.* AGU Fall Meeting, New Orleans, LA, USA, 15–19 December 2025.	2025
4.	*Using satellite data to characterize air pollution and health in cities and countries.* EDF, Climate and Clean Air Coalition and UNEP Workshop for Clean Air Solutions in Latin America and the Caribbean, Bogotá, Colombia, 27 April 2023.	2023

RESEARCH SUPERVISION

I have supervised 20 researchers from BSc to postdoctoral level across different countries and disciplinary contexts (geography, public health and computer science), which has made me an adaptable supervisor. My supervision philosophy is educative mentoring, in which I guide my mentees through shared investigation and joint problem-solving.

Postdoctoral

Lucas Ngo, Cardiff University (**Primary Supervisor**) 2026–present

PhD

Diya Wang, Cardiff University (**Primary Supervisor**) 2026–present

Feichi Hu, Cardiff University (**Co-Supervisor** with L. Zhuo as Primary) 2025–present

PhD examination and progression review

Xin Wang, Cardiff University (**Internal Review Chair**) 2026–present
Patrick Wiecko, CU Boulder (**External Review Panel Member**) 2026–present

MSc, MSci and MPH

Mia Kate Petts, MSc, Cardiff University (**Primary Supervisor**) 2026–present
Mohammed Fahad H Alsuwayh, MSc, Cardiff University (**Primary Supervisor**) 2026–present
Kyla Camama, MSci, Cardiff University (**Primary Supervisor**) 2025–2026
Katie O'Donnell, MPH, GWU (**Co-Supervisor** with S. Anenberg as Primary) 2023–2024
Erin Campbell, MPH, GWU (**Co-Supervisor** with S. Anenberg as Primary) 2023–2024

BSc

10 BSc Geography dissertations, Cardiff University (**Primary Supervisor**) 2025–present
Olivia Paquette, GWU (**Co-Supervisor** with D. Goldberg as Primary) 2024–2025
Mohammed Alwakeel, CU Boulder (**Co-Supervisor** with D. Henze as Primary) 2019–2021

TEACHING

My pedagogical philosophy is inclusive social constructivism, which I adapt to different educative contexts. In large lectures (more than 100 students) I apply this through think-pair-share exercises and interactive crowd polling (Mentimeter). For postgraduate seminar-style teaching, I apply it through Socratic dialogue and interactive social exercises such as a mock COP. In my practical data analysis computer labs, I demonstrate coding and GIS exercises at the front of the classroom for a fraction of the time and then guide students through analytical problems in individual and group discussions. At Cardiff University my teaching has been nominated by my students for an Enriching Student Life Award, and I completed the teaching fellowship course to achieve FHEA status and develop my pedagogy.

Cardiff University, School of Physical, Chemical and Environmental Sciences

The Ocean-Atmosphere System, Year 2 (**Module Leader**) 2025–present
GIS, Maps and Analytical Skills, Year 1 (**Lecturer**) 2025–present
Digital Fieldwork Workshop, Year 1 (**Lecturer**) 2025–present
Climate Change Adaptation and Resilience, MSc (**Lecturer**) 2026–present
Geography Tutorials, Years 1–3 (**Personal Tutor**) 2025–present

George Washington University, Department of Environmental and Occupational Health

Global Climate Change and Air Pollution, MPH (**Lecturer**) 2024

University of Colorado Boulder, Department of Mechanical Engineering

PhD Preliminary Exam Review, PhD (**Mentor**) 2020
Numerical Methods for Engineers, Year 1 (**Lead Teaching Assistant**) 2018–2019

LEADERSHIP AND SERVICE

School leadership and committees, Cardiff University

Equality, Diversity, Inclusion and Anti-Racism Lead 2026–present
My primary administrative role, in which I lead EDI&A strategy and culture, advise on equality and inclusion matters, and oversee delivery of chartermark activity for a School of approximately 2,000 students and 300 staff.
Member, Future Generations Engagement Fund Review Panel 2026
Member, Staff Promotions Panel 2026
Member, Extenuating Circumstances Panel 2026
Director of EDI, School of Earth and Environmental Sciences 2025
Member, Research Ethics Committee 2025–present
Member, Research Leave Panel 2025
Contributor to induction for Year 1 students 2025–present

Editorial roles

Associate Editor, *Air Quality, Atmosphere & Health* 2026–present
Associate Editor, *Frontiers in Environmental Health* (Environmental Epidemiology) 2025–2026

Peer review for journals (46)

Atmospheric Chemistry and Physics (1); *Communications Earth & Environment* (2); *Discover Atmospheres* (1); *Discover Cities* (1); *Earth's Future* (1); *Earth System Science Data* (1); *Elementa: Science of the Anthropocene* (1); *Environmental Monitoring & Assessment* (1); *Environmental Research Letters* (2); *Environmental Science & Technology* (3); *Environmental Science & Technology Air* (1); *Environmental Science Policy Research* (1); *GeoHealth* (3); *Geoscience Letters* (1); *Health Data Science* (1); *Humanities & Social Science Communications* (1); *International Geoscience and Remote Sensing Symposium* (6); *Journal of the Air and Waste Management Association* (1); *Lancet Planetary Health* (1); *Nature* (1); *Nature Aging* (1); *Nature Cities* (1); *Nature Communications* (1); *Nature Food* (1); *Nature Health* (4); *Nature Sustainability* (1); *npj Clean Air* (1); *npj Climate & Atmospheric Science* (1); *Open Research Europe* (1); *PLOS* (1); *PNAS* (1); *Scientific Reports* (1)

Peer review for proposals (2)

Wellcome Trust Discretionary Award, expert review (2)

Conference organisation

Session convenor on air quality, climate, health and equity, EGU General Assembly 2026
Session convenor on air quality, climate, health and equity, EGU General Assembly 2025
Conference coordinator, Community Modeling and Analysis System (CMAS) conference 2017

PROFESSIONAL MEMBERSHIPS AND NETWORKS

EQUATOR Mentor 2025–present
TOAR-II Health Team (Tropospheric Ozone Assessment Report, Phase II) 2025–present
COST Action CliMent (CA23113) 2025–present
Global Burden of Disease Study Collaborator 2024–present
European Geosciences Union 2023–present
American Geophysical Union 2018–present

KNOWLEDGE EXCHANGE, PUBLIC ENGAGEMENT AND OUTREACH

Reports and policy briefs

Jin, L., Benoit, J., Walzer-Goldfeld, S., **Nawaz, M. O.**, Wiecko, P., Sen, A., Miller, J. *Health benefits of zero-emission transport through 2050*. Report, International Council on Clean Transportation, 29 June 2026, with national and state fact sheets (United States, California, New York, Illinois, India) and a Bahasa Indonesia edition, link here 2026

Decker, T., Meng, Z., Comer, B. *Health and air pollution benefits of a global 0.1% fuel sulfur limit on marine fuels*. Policy brief, International Council on Clean Transportation, 8 July 2025; contributed the health impact analysis (acknowledged), link here 2025

Press releases

International Council on Clean Transportation, *Vehicle pollution cuts one life short every 45 seconds: ambitious electrification could save 8.8 million by 2050*, 29 June 2026, link here 2026

International Council on Clean Transportation, *Vehicle pollution responsible for 41,800+ deaths, 23,100+ asthma cases in the U.S. every year*, 29 June 2026, link here 2026

Cardiff University, *Climate policy must consider cross-border pollution "exchanges" to address inequality and achieve health benefits, research finds*, 12 February 2026, link here 2026

International Council on Clean Transportation, *Policies targeting road transport emissions could save 1.9 million lives globally by 2040, new study finds*, 19 May 2025, link here 2025

Cardiff University, *Policies targeting road transport emissions could save 1.9 million lives globally by 2040, new study finds*, 19 May 2025, link here 2025

Events

Expert presenter and panel member, NERC and Sciencewise public dialogue, *Exploring public priorities for the future of atmospheric science in the UK*	2026
Staff volunteer, School of Earth and Environmental Sciences Open Day, Cardiff University	2025

Selected media features

BBC News Radio Wales, *Cleaner air in steel town a year since furnace shut*, 23 December 2025, link here	2025
WXPR, *The health impacts of tailpipe emissions get a fresh look*, 11 August 2025, link here	2025
Eos Earth and Space Science News, *The Global Health Benefits of Going Net Zero*, 14 February 2023, link here	2023
Revista Pesquisa FAPESP, *Queimadas na Amazônia aumentam internações (Fires in the Amazon increase hospitalisations)*, 18 September 2020, link here	2020

APPENDIX: FULL LIST OF PUBLICATIONS AND PRESENTATIONS

In press (1)

1. Kerr, G. H., **Nawaz, M. O.**, Anenberg, S. C., Anthoff, D., Burton, C., Carter, T. S., Henze, D. K., Kelley, D. I., Kingdon, C., O'Dell, K., Prest, B. C., Cromar, K. R. (2026). Climate-driven surges in public health damages from wildland fire-sourced pollution through 2100. *Nature Health*, accepted.

Peer-reviewed publications (27)

1. Kerr, G. H., Amegah, K., Chen, L., … **Nawaz, M. O.**, … Yan, Y. (2026). Tropospheric Ozone Assessment Report II: Changing health impacts of global ozone exposure. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 384, 20250268.
2. Goldberg, D. L., Johnson, J., Shah, T., **Nawaz, M. O.**, Kerr, G. H., Chen, L., Huber, D., Judd, L. (2026). Contrasting the TEMPO and TROPOMI views of tropospheric NO_2: Implications for diurnal variability and model evaluation. *Journal of Geophysical Research: Atmospheres*, 131, e2026JD046905.
3. Huber, D. E., Kerr, G. H., **Nawaz, M. O.**, Runkel, S., Anenberg, S. C., Goldberg, D. L. (2026). Global NO_2 changes between 2019 and 2024 as observed by TROPOMI in urban areas and emerging hotspots. *Atmospheric Chemistry and Physics*, 26, 3783–3803.
4. **Nawaz, M. O.**, Henze, D. K. (2026). National climate action can ameliorate, perpetuate, or exacerbate international air pollution inequalities. *Nature Communications*, doi:10.1038/s41467-026-68827-0.
5. Siu, T. K., Goldberg, D. L., Kerr, G. H., Chen, L., **Nawaz, M. O.**, Chang, R. Y.-W., Fong, K. C. (2025). Tropospheric NO_2 patterns in eastern Canada using the first year of TEMPO observations. *Journal of Geophysical Research: Atmospheres*, 130, e2025JD044757.
6. Goldberg, D. L., **Nawaz, M. O.**, Lyu, C., He, J., Carlton, A. G., Kondragunta, S., Anenberg, S. C. (2025). Clear-sky and cloudy-sky differences in NO_2 concentrations over the United States: implications for satellite measurement applications. *Atmospheric Chemistry and Physics*, 25, 16287–16302.
7. **Nawaz, M. O.**, Huber, D. E., Kerr, G. H., Judd, L. M., Acker, S. J., Goldberg, D. L. (2025). A comparative analysis of TEMPO NO_2 remote sensing with surface-level monitoring through diurnal and seasonal trends, meteorology, and monitor characteristics. *Journal of Geophysical Research: Atmospheres*, 130, e2025JD043923.
8. Hay, S. I., Ong, K. L., Santomauro, D. F., … **Nawaz, M. O.**, … Gakidou, E. (2025). Burden of 375 diseases and injuries, risk-attributable burden of 88 risk factors, and healthy life expectancy in 204 countries and territories, including 660 subnational locations, 1990–2023: a systematic analysis for the Global Burden of Disease Study 2023. *The Lancet*, 406, 1873–1922.
9. Jin, L., Benoit, J., **Nawaz, M. O.**, Rodrigues, P. F., Wiecko, P., Miller, J., Alvarez, G., Henze, D. K., Osipova, L., Anenberg, S. (2025). Global health benefits of policies to reduce on-road vehicle pollution through 2040. *Environmental Research Letters*, 20, 064005.
10. Wiecko, P., Henze, D. K., **Nawaz, M. O.** (2025). Sector-, season-, and country-specific NO_2-associated health benefits from NO_x emission reductions. *ACS ES&T Air*, 2, 700–709.
11. **Nawaz, M. O.**, Goldberg, D. L., Kerr, G. H., Anenberg, S. C. (2025). TROPOMI satellite data reshape NO_2 air pollution land-use regression modeling capabilities in the United States. *ACS ES&T Air*, 2, 187–200.
12. Dyer, G. M. C., Khomenko, S., Adlakha, D., … **Nawaz, M. O.**, … Nieuwenhuijsen, M. (2024). Commentary: A road map for future data-driven urban planning and environmental health research. *Cities*, 155, 105340.
13. Goldberg, D. L., de Foy, B., **Nawaz, M. O.**, Johnson, J., Yarwood, G., Judd, L. (2024). Quantifying NO_x emission sources in Houston, Texas using remote sensing aircraft measurements and source apportionment regression models. *ACS ES&T Air*, 1, 1391–1401.
14. Dyer, G. M. C., Khomenko, S., Adlakha, D., … **Nawaz, M. O.**, … Nieuwenhuijsen, M. (2024). Exploring the nexus of urban form, transport, environment and health in large-scale urban studies: A state-of-the-art scoping review. *Environmental Research*, 257, 119324.
15. **Nawaz, M. O.**, Johnson, J., Yarwood, G., de Foy, B., Judd, L., Goldberg, D. L. (2024). An intercomparison of satellite, airborne, and ground-level observations with WRF–CAMx simulations of NO_2 columns over Houston, Texas, during the September 2021 TRACER-AQ campaign. *Atmospheric Chemistry and Physics*, 24, 6719–6741.
16. Choi, J., Henze, D. K., **Nawaz, M. O.**, Malley, C. S. (2024). Source attribution of health burdens from ambient $PM_{2.5}$, O_3, and NO_2 exposure for assessment of South Korean national emission control scenarios by 2050. *GeoHealth*, 8, e2024GH001042.
17. Gu, Y., Henze, D. K., **Nawaz, M. O.**, Wagner, U. J. (2023). Response of the ozone-related health burden in Europe to changes in local anthropogenic emissions of ozone precursors. *Environmental Research Letters*, 18, 114034.
18. **Nawaz, M. O.**, Henze, D. K., Huneeus, N. J., Osses, M., Álamos, N., Opazo, M. A., Gallardo, L. (2023). Sources of air pollution health impacts and co-benefits of carbon neutrality in Santiago, Chile. *Journal of Geophysical Research: Atmospheres*, 128, e2023JD038808.
19. Jo, D. S., Nault, B. A., Tilmes, S., Gettelman, A., McCluskey, C. S., Hodzic, A., Henze, D. K., **Nawaz, M. O.**, Fung, K. M., Jimenez, J. L. (2023). Global health and climate effects of organic aerosols from different sources. *Environmental Science & Technology*, 57, 13793–13807.

20. Gu, Y., Henze, D. K., **Nawaz, M. O.**, Cao, H., Wagner, U. J. (2023). Sources of $PM_{2.5}$-associated health risks in Europe and corresponding emission-induced changes during 2005–2015. *GeoHealth*, 7, e2022GH000767.
21. **Nawaz, M. O.**, Henze, D. K., Anenberg, S. C., Ahn, D. Y., Goldberg, D. L., Tessum, C. W., Chafe, Z. A. (2023). Sources of air pollution-related health impacts and benefits of radially applied transportation policies in 14 US cities. *Frontiers in Sustainable Cities*, 5.
22. **Nawaz, M. O.**, Henze, D. K., Anenberg, S. C., Braun, C., Miller, J., Pronk, E. (2023). A source apportionment and emission scenario assessment of $PM_{2.5}$- and O_3-related health impacts in G20 countries. *GeoHealth*, 7, e2022GH000713.
23. **Nawaz, M. O.**, Henze, D. K., Harkins, C., Cao, H., Nault, B., Jo, D., Jimenez, J., Anenberg, S. C., Goldberg, D. L., Qu, Z. (2021). Impacts of sectoral, regional, species, and day-specific emissions on air pollution and public health in Washington, DC. *Elementa: Science of the Anthropocene*, 9, 00043.
24. Cao, H., Henze, D. K., Cady-Pereira, K., McDonald, B. C., Harkins, C., Sun, K., Bowman, K. W., Fu, T.-M., **Nawaz, M. O.** (2021). COVID-19 lockdowns afford the first satellite-based confirmation that vehicles are an under-recognized source of urban NH_3 pollution in Los Angeles. *Environmental Science & Technology Letters*, 9, 3–9.
25. Nault, B. A., Jo, D. S., McDonald, B. C., … **Nawaz, M. O.**, … Jimenez, J. L. (2021). Secondary organic aerosols from anthropogenic volatile organic compounds contribute substantially to air pollution mortality. *Atmospheric Chemistry and Physics*, 21, 11201–11224.
26. Malley, C. S., Hicks, W. K., Kulyenstierna, J. C. I., … **Nawaz, M. O.**, … Robinson, T. P. (2021). Integrated assessment of global climate, air pollution, and dietary, malnutrition and obesity health impacts of food production and consumption between 2014 and 2018. *Environmental Research Communications*, 3, 075001.
27. **Nawaz, M. O.**, Henze, D. K. (2020). Premature deaths in Brazil associated with long-term exposure to $PM_{2.5}$ from Amazon fires between 2016 and 2019. *GeoHealth*, 4, e2020GH000268.

Manuscripts under review (8)

1. Graffy, P. M., … **Nawaz, M. O.**, … et al. The association of air pollution with death and deterioration on the United States transplant waiting list.
2. Graffy, P. M., … **Nawaz, M. O.**, … et al. Cumulative air pollution exposure increases the severity of acute respiratory failure: a U.S. multicenter study.
3. Graffy, P. M., … **Nawaz, M. O.**, … et al. Respiratory trajectory phenotypes and air pollution-associated outcomes in critically ill patients with lung cancer in the United States.
4. Gu, Y., Henze, D. K., **Nawaz, M. O.**, Zhang, L., Liao, H. Non-agricultural NH_3 emissions disproportionately contribute to urban $PM_{2.5}$ health burdens in China.
5. Hu, F., Zhuo, L., **Nawaz, M. O.**, Wen, Z., Han, D. Improving TROPOMI methane plume candidate screening: gradient boosting, chemical context, and image-based approaches.
6. Kim, S. Y., Kerr, G. H., **Nawaz, M. O.**, Anenberg, S. C. Fine-scale spatiotemporal patterns of NO_2 pollution and associated mortality burdens across the continental United States.
7. Lang, V. A., Partha, D., Goldberg, D. L., … **Nawaz, M. O.**, … Horton, D. E. A framework for augmenting medium- and heavy-duty vehicle emissions within the National Emissions Inventory for hyperlocal air quality assessments.
8. Morris, S. T., O'Neill, B. C., Msangi, S., **Nawaz, M. O.**, Parker, N., Rao, N., Van Vuuren, D. Modeling the human well-being dimensions of global change: priorities and challenges for research to inform decision-making.

First author oral presentations (13)

1. *Modeling approaches for inferring surface-level NO_2 using remote sensing.* Cardiff University School of Earth and Environmental Sciences Seminar Series, Cardiff, UK, 17 February 2026. — 2026
2. *Exploring the role of climate action in transboundary air pollution inequality using GEOS-Chem adjoint sensitivities.* EGU General Assembly, Vienna, Austria, 29 April 2025. — 2025
3. *Development of a land-use regression of hourly surface NO_2 in preparation for GeoXO atmospheric composition data.* EGU General Assembly, Vienna, Austria, 17 April 2024. — 2024
4. *Value of GeoXO atmospheric composition data for estimating air pollution-related health impacts.* AMS Annual Meeting, Baltimore, MD, USA, 30 January 2024. — 2024
5. *Investigating climate co-benefits using GEOS-Chem adjoint sensitivities.* Second GEOS-Chem Europe Meeting, London, UK, 15 August 2023. — 2023
6. *Regional vs local sources of municipal air pollution-related health impacts.* AMS Annual Meeting, Denver, CO, USA (presented by D. K. Henze), 10 January 2023. — 2023
7. *Developing an interactive tool for characterizing the air pollution-related health impacts in Los Angeles, CA associated with different proposed emission scenarios.* Earth Science Information Partners Meeting, Pittsburgh, PA, USA (virtual), 19 July 2022. — 2022
8. *Leveraging satellite-derived data in GEOS-Chem adjoint simulations to characterize the sources of $PM_{2.5}$-, O_3-, and NO_2-related health impacts at multiple spatial scales.* 10th International GEOS-Chem Meeting, St Louis, MO, USA (virtual), 9 June 2022. — 2022

9. *Characterizing the sources of air pollution at the urban and country scale: case studies in Santiago, Chile and G20 countries.* Graduate Engineering Annual Research and Recruitment Symposium, Boulder, CO, USA, 17 February 2022. 2022
10. *Comparing domestic and extra-regional contributions to pollutant exposures and health impacts in G20 countries through a novel adjoint modeling approach.* AGU Fall Meeting, New Orleans, LA, USA (virtual), 15 December 2021. 2021
11. *Characterizing the regional, sectoral and species-specific sources of pollution exposure and its associated health impacts in urban environments: case studies in Washington, DC and Santiago, Chile.* AGU Fall Meeting (virtual), 14 December 2020. 2020
12. *Premature deaths in Brazil associated with long-term exposure to $PM_{2.5}$ from Amazon fires and development of a nested South American domain for the GEOS-Chem adjoint.* 19th GEIA Conference (virtual), 23 June 2020. 2020
13. *Source attribution of $PM_{2.5}$ from sensitivity analyses in the GEOS-Chem adjoint model.* Young Scientists Symposium on Atmospheric Research, Fort Collins, CO, USA, 25 October 2019. 2019

First author poster presentations (15)

1. *Estimating the air pollutant-attributable health burden of the oil and gas sector in Mexico using a TROPOMI flux-divergence approach.* EGU General Assembly, Vienna, Austria, 7 May 2026. 2026
2. *What does low-earth orbiting, geostationary, and airborne remote sensing reveal about surface NO_2?* UK Atmospheric Chemistry Conference, York, UK, 9 September 2025. 2025
3. *Characterizing the air quality and health impacts from oil and gas emissions in Mexico using GCHP.* GEOS-Chem Europe Meeting 3, London, UK, 19 August 2025. 2025
4. *What can TROPOMI and TEMPO remote sensing reveal about seasonal and diurnal trends in surface-level NO_2?* AGU Fall Meeting, Washington, DC, USA, 13 December 2024. 2024
5. *Leveraging satellite-derived data and air quality modeling to characterize source profiles of climate co-benefits at the urban and country scale.* AGU Fall Meeting, Chicago, IL, USA, 12 December 2022. 2022
6. *Health benefits of decreases in $PM_{2.5}$ and ozone in the United States, 1990–2016.* NASA Health and Air Quality Applied Sciences Team Final Showcase (virtual), 21 July 2020. 2020
7. *Source attribution of $PM_{2.5}$ and O_3 concentrations and health outcomes from 2010 and 2011 in Washington, DC using sensitivity analyses in the GEOS-Chem adjoint model.* AGU Fall Meeting, San Francisco, CA, USA, 9 December 2019. 2019
8. *Source attribution of climate and health impacts from aerosols.* 9th International GEOS-Chem Meeting, Cambridge, MA, USA, 6 May 2019. 2019
9. *The use of adjoint modeling to assess the sources of air pollution and its associated health impacts.* Graduate Engineering Annual Research and Recruitment Symposium, Boulder, CO, USA, 21 February 2019. 2019
10. *Source attribution of climate and health impacts from aerosols.* AGU Fall Meeting, Washington, DC, USA, 14 December 2018. 2018
11. *Health benefits of decreases in $PM_{2.5}$ and ozone in the United States, 1990–2015.* NASA Health and Air Quality Applied Sciences Team Meeting, Madison, WI, USA, 16 July 2018. 2018
12. *Health benefits of decreases in $PM_{2.5}$ and ozone in the United States, 1990–2015.* Climate Change and Resilience Symposium, Chapel Hill, NC, USA, 20 April 2018. 2018
13. *Health benefits of decreases in $PM_{2.5}$ and ozone in the United States from 1990 to 2015.* AGU Fall Meeting, New Orleans, LA, USA, 11 December 2017. 2017
14. *Health benefits of decreases in $PM_{2.5}$ and ozone in the United States from 1990 to 2015.* Community Modeling and Analysis System Conference, Chapel Hill, NC, USA, 23 October 2017. 2017
15. *Impact of regional ozone precursor emissions on global ozone burden.* Celebration of Undergraduate Research, Chapel Hill, NC, USA, 12 April 2017. 2017